Abbas Taqvi

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Professional Summary

- Data Engineer and M.S. Data Science student at Milano-Bicocca, with a founding-engineer background building lakehouse and streaming platforms (Kafka, Spark, Trino, Airflow, AWS) over 30B+ records. Built a Rust ETL framework (10× faster) and a LangGraph Text2SQL agent that cut analyst turnaround by 41%. Seeking a data engineering internship in Italy.

Education

- **Università degli Studi di Milano-Bicocca** — Milan, Italy
  Master of Science in Data Science — *Oct 2025 – Present*
- **Habib University** — Karachi, Pakistan
  Bachelor of Science in Computer Science; GPA: 3.31; 100% merit scholarship (HU-TOPS) — *Aug 2020 – May 2024*

Experience

- **Logiciel Services LLC** — United States (Remote)
  Software Engineer (Founding) — *May 2024 – Nov 2025*
  - **Data Platform**: Built the entire backend data platform end-to-end: ingestion (Kafka), processing (Spark/Scala), storage (S3 lakehouse), query engine (Trino) and orchestration (Airflow). Simplified a 12-component architecture to 5, housing 30B+ records queryable in <1s.
  - **Ferroflow**: Rust ETL framework with composable processor/pipeline abstractions, replacing a legacy .NET monolith: 500 min → 50 min daily (10× speedup), saving 400 EC2-minutes per run.
  - **Text2SQL Chatbot**: Multi-agent chatbot (Python, LangGraph, Trino) that translates natural language into SQL over the data lake via REST API, reducing analyst turnaround by 41%.
- **Pullflow Inc.** — San Jose, CA (Remote)
  DevOps Intern — *Jul 2023 – Aug 2023*
  - **Kubernetes Migration**: Migrated production services from AWS EKS to on-premise Kubernetes (MicroK8s): wrote Dockerfiles, Helm charts and a Traefik reverse-proxy config, with a zero-downtime cutover.
- **United Bank Limited** — Karachi, Pakistan
  Data Engineering Intern — *May 2023 – Jun 2023*
  - **Streaming Pipeline**: Built a cross-region streaming pipeline (Informatica) and validated production AI systems against new regional schemas across 3 countries.
- **ImproData** — Pinedale, WY (Remote)
  Software Engineer Intern — *Jan 2021 – May 2022*
  - **Data Pipelines**: Built pipelines on Azure Databricks (PySpark) and automated ingestion via GCP Cloud Functions (JSON → GCS → BigQuery), eliminating 3 hours of daily manual work.

Projects

- **Kuza: End-to-End Data Platform**: Spark (Scala) ingestion for 375+ daily sessions, Kafka streaming, S3 lakehouse, Trino and Airflow with DAGs auto-generated from YAML (Scala, Spark, Kafka, Airflow, Trino, AWS, Docker).
- **Yohsin: Zero-Copy Object Serializer**: Rust serializer using memory-mapped files, O(1) random access and a write-ahead log for sub-50 ms recovery; sustained 1M records/s (2.25 GB/s). Built in 24 hours at a hackathon (Rust, tokio, mmap).

Skills

- **Languages**: Python, Rust, Scala, C/C++ (CUDA), Go, SQL, JavaScript
- **Data & Cloud**: Spark, Kafka, Airflow, Trino, Iceberg, Databricks, AWS (S3, EKS, EC2, Glue), Azure, GCP
- **Systems & Infrastructure**: Linux, Docker, Kubernetes (EKS, MicroK8s), Traefik, REST APIs, gRPC, Git
- **AI / ML**: PyTorch, JAX, TensorFlow, LangChain, LangGraph, Pydantic-AI
- **Databases**: PostgreSQL, MongoDB, Neo4j, Qdrant, Weaviate, Amazon RDS
- **Spoken Languages**: English (C1), Italian (A2)

Achievements & Test Scores

- **Competitions**: ICPC Asia Topi Regionals 2022, 19th place; Game of Codes 5.0, 1st place
- **Tests & Certification**: IELTS 8.0/9.0; GRE Quantitative 165/170; NVIDIA Accelerated Computing (CUDA C/C++)

Autorizzo il trattamento dei dati personali ai sensi del Regolamento UE 2016/679 (GDPR).